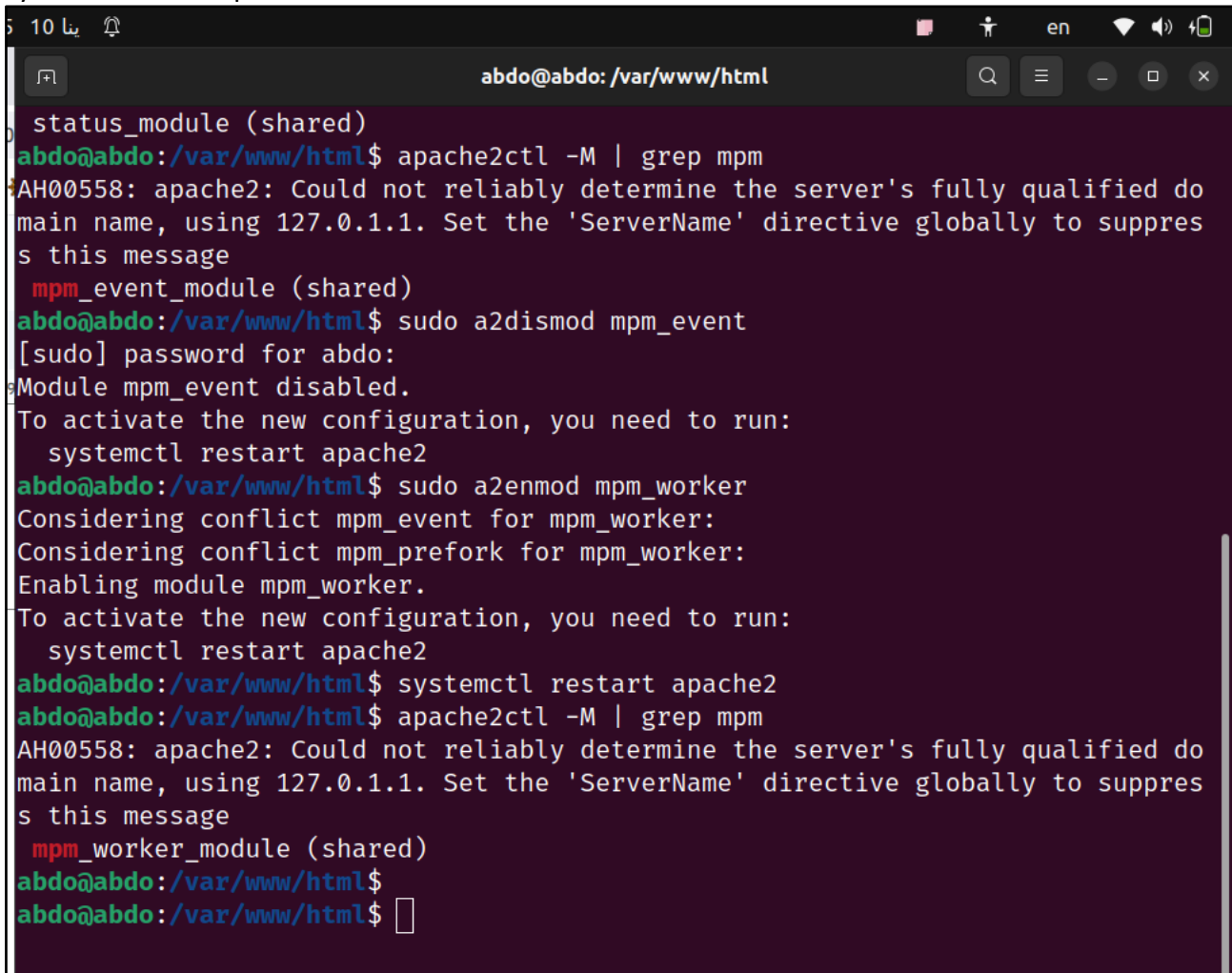


1. Configure apache to use Worker MPM

```
sudo a2dismod mpm_event
sudo a2enmod mpm_worker
systemctl restart apache2
```

A terminal window with a dark purple background and light green text. The window title is 'abdo@abdo: /var/www/html'. The terminal shows the following commands and output:

```
status_module (shared)
abdo@abdo:/var/www/html$ apache2ctl -M | grep mpm
AH00558: apache2: Could not reliably determine the server's fully qualified do
main name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress
s this message
mpm_event_module (shared)
abdo@abdo:/var/www/html$ sudo a2dismod mpm_event
[sudo] password for abdo:
Module mpm_event disabled.
To activate the new configuration, you need to run:
  systemctl restart apache2
abdo@abdo:/var/www/html$ sudo a2enmod mpm_worker
Considering conflict mpm_event for mpm_worker:
Considering conflict mpm_prefork for mpm_worker:
Enabling module mpm_worker.
To activate the new configuration, you need to run:
  systemctl restart apache2
abdo@abdo:/var/www/html$ systemctl restart apache2
abdo@abdo:/var/www/html$ apache2ctl -M | grep mpm
AH00558: apache2: Could not reliably determine the server's fully qualified do
main name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress
s this message
mpm_worker_module (shared)
abdo@abdo:/var/www/html$
abdo@abdo:/var/www/html$
```

2. Check the status of rewrite_mod and enable it if it is enabled.

```
apachectl -M | grep rewrite
sudo a2enmod rewrite
sudo systemctl restart apache2
```

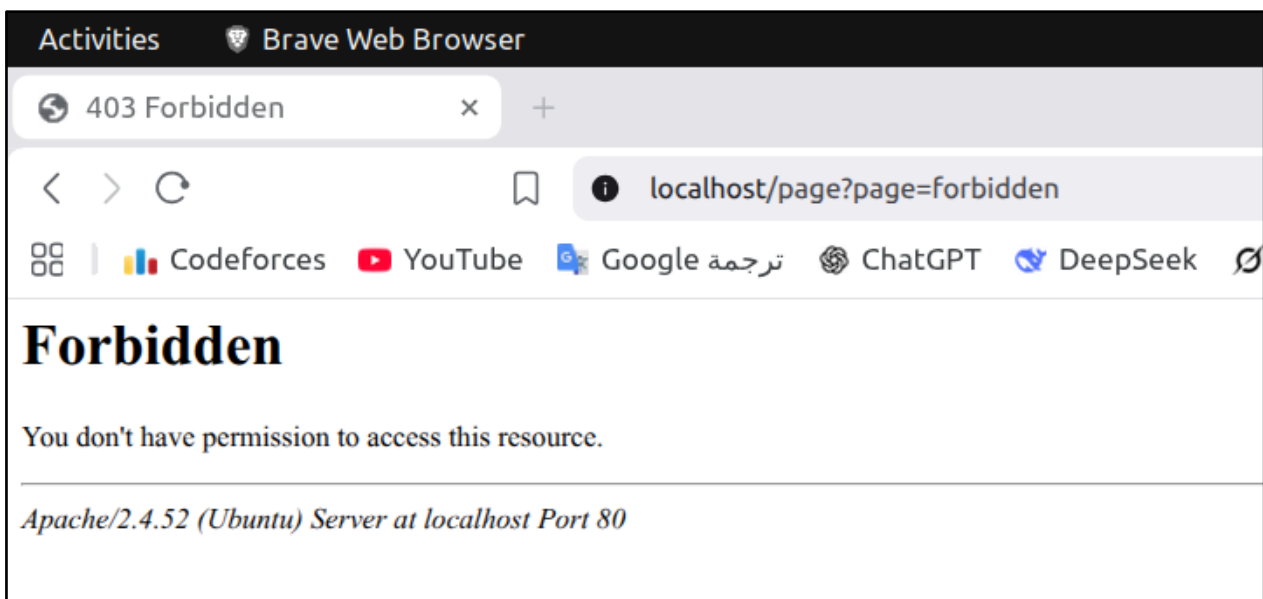
```
3 10 10
abdo@abdo: /var/www/html
abdo@abdo:/var/www/html$ apachectl -M | grep rewrite
AH00558: apache2: Could not reliably determine the server's fully qualified do
main name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress
s this message
abdo@abdo:/var/www/html$ sudo a2enmod rewrite
Enabling module rewrite.
To activate the new configuration, you need to run:
systemctl restart apache2
abdo@abdo:/var/www/html$ sudo systemctl restart apache2
abdo@abdo:/var/www/html$ apachectl -M | grep rewrite
AH00558: apache2: Could not reliably determine the server's fully qualified do
main name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress
s this message
rewrite_module (shared)
abdo@abdo:/var/www/html$
```

3. Write RewriteCond and RewriteRules for:

Sudo nano /etc/apache2/sites-available/000-default.conf

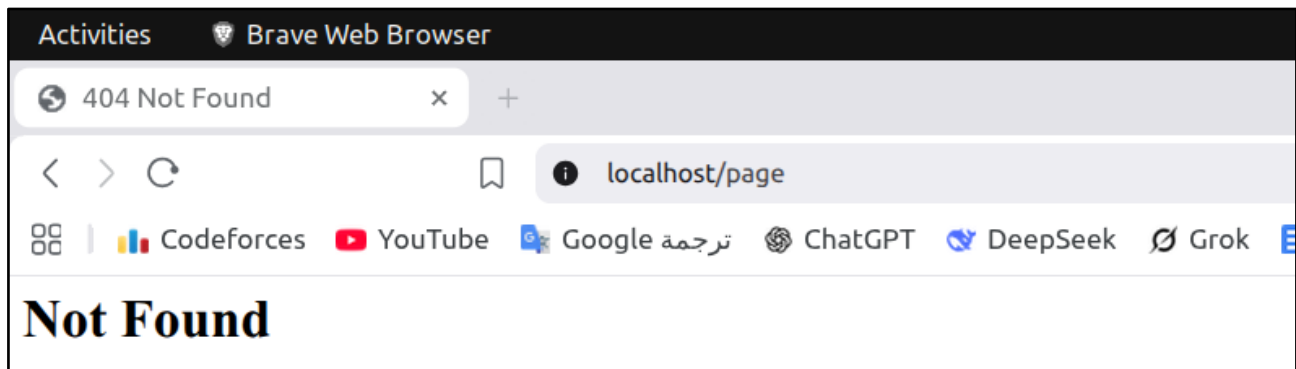
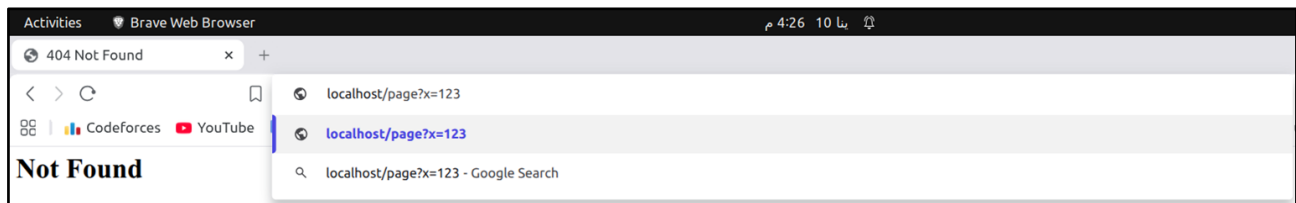
a. Deny access to `http://localhost/page?queryString` if queryString contains the string forbidden.

```
#RewriteCond %{QUERY_STRING} forbidden
#RewriteRule ^page$ - [F]
```



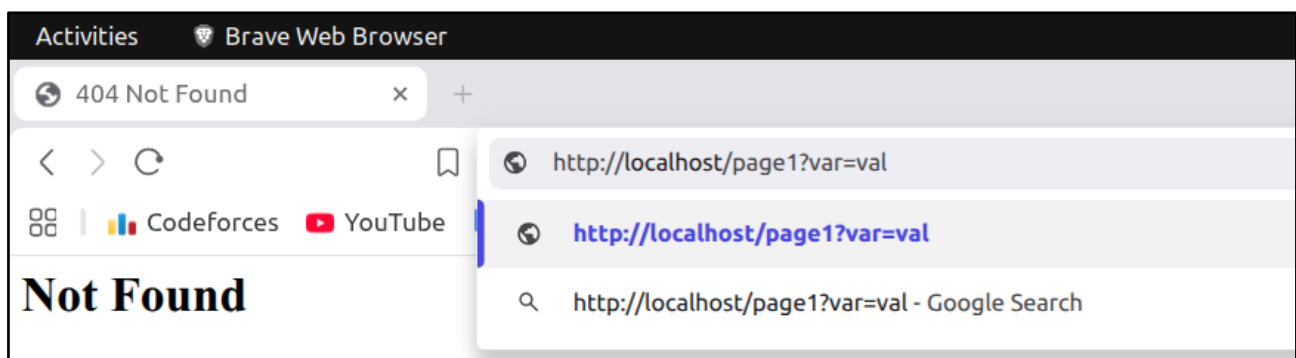
b. Remove the Query String

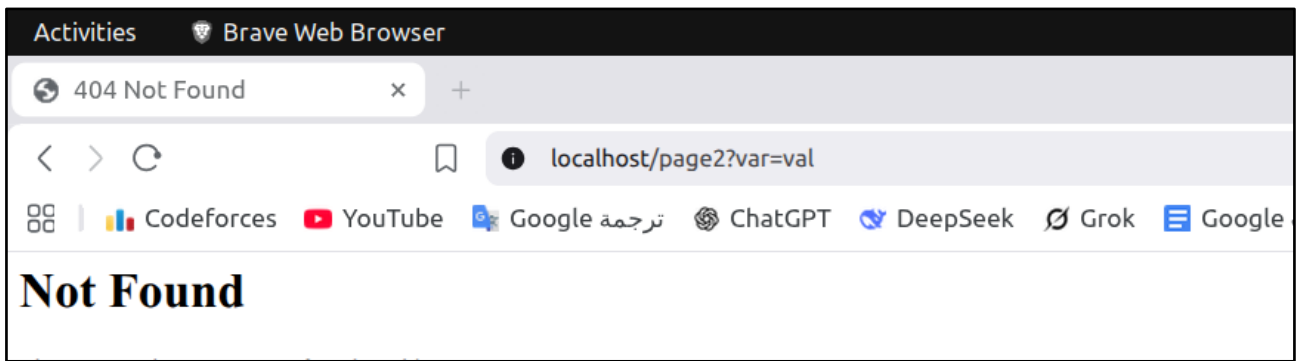
```
#RewriteCond %{QUERY_STRING} .  
#RewriteRule ^(.*)$ /$1? [R]
```



c. Rewrite URLs like `http://localhost/page1?var=val` to `http://localhost/page2?var=val` but do not rewrite if `val` isn't present.

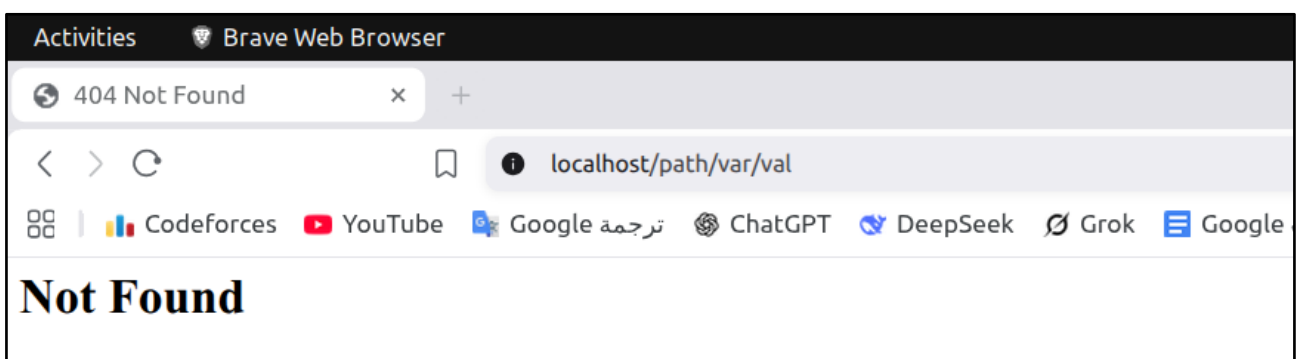
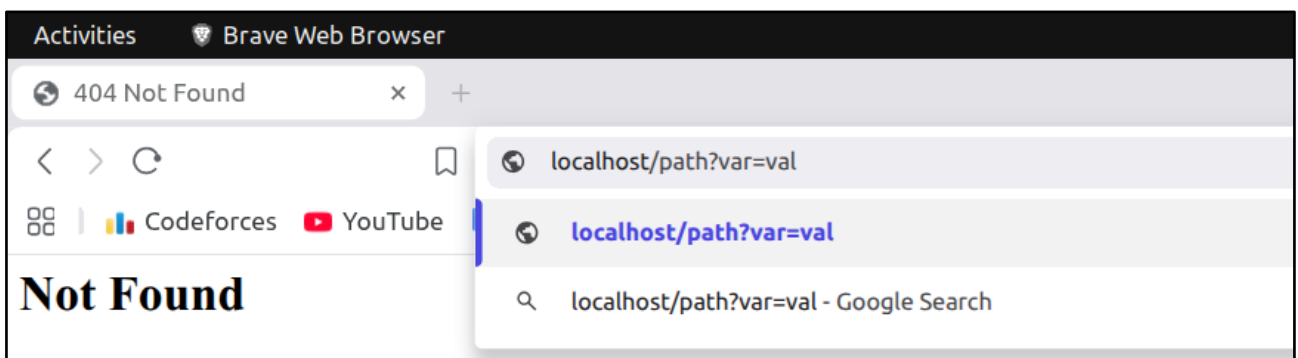
```
#RewriteCond %{QUERY_STRING} ^var=([^&]+)  
#RewriteRule ^page1$ /page2?%{QUERY_STRING} [R,L]
```





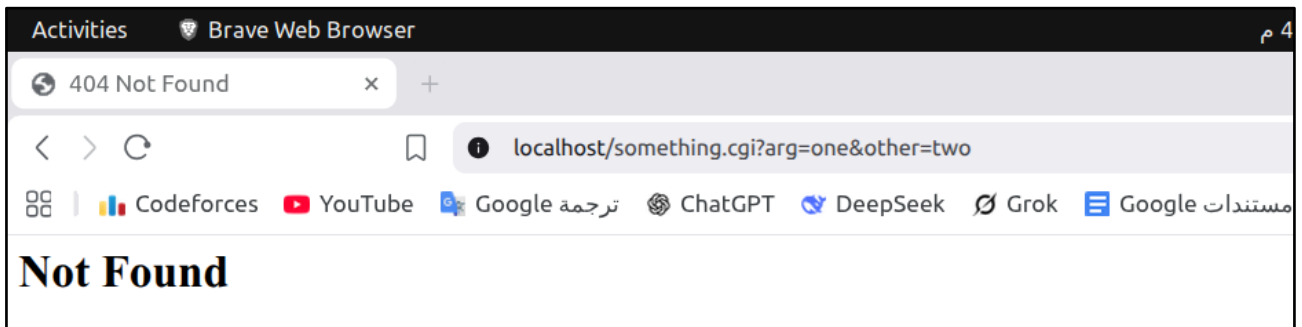
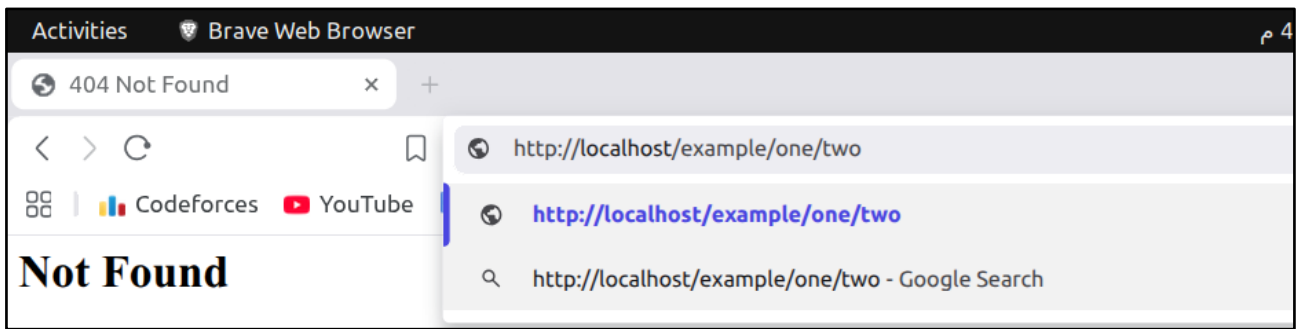
d. Take a URL of the form <http://localhost/path?var=val> and transform it into <http://localhost/path/var/val>.

```
#RewriteCond %{QUERY_STRING} ^var=([^&]+)
#RewriteRule ^path$ /path/var/%1? [R=302,L]
```



e. Map <http://localhost/example/one/two> to <http://localhost/something.cgi?arg=one&other=two>

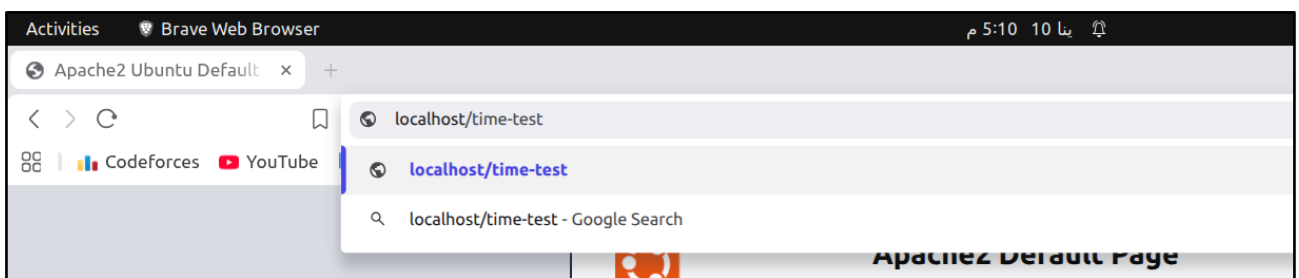
```
RewriteRule ^example/([^/]+)/([^/]+$) /something.cgi?arg=$1&other=$2 [R=302,L]
```

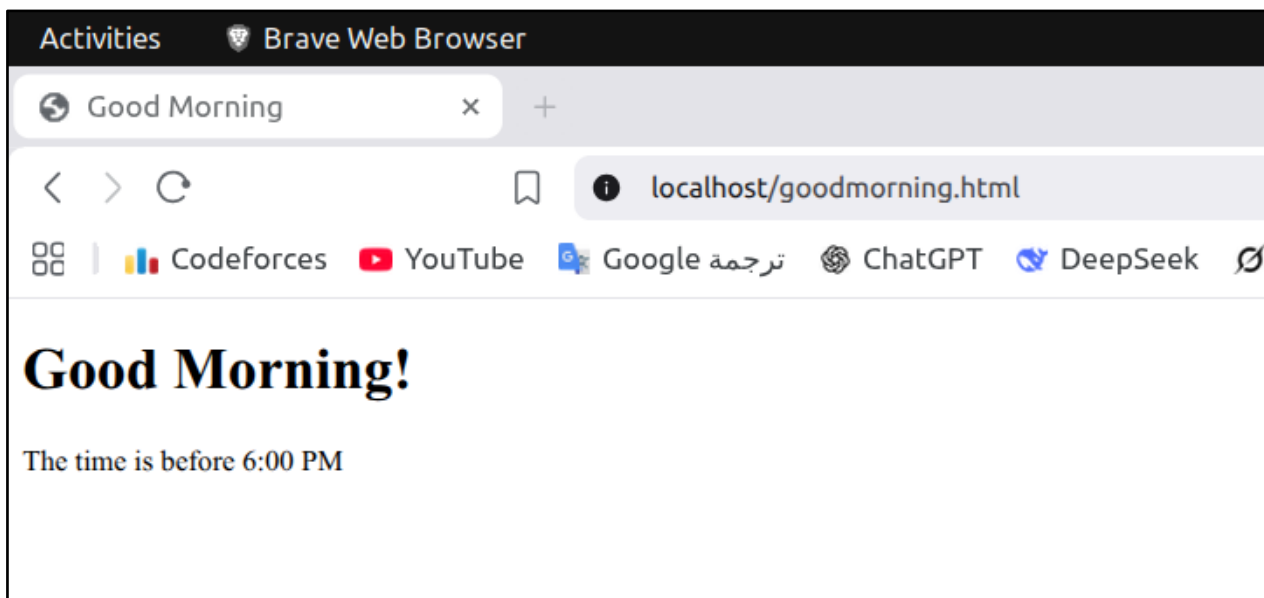


f. (Bonus) Display a `goodevening.html` page if the time is after 6:00 PM and `goodmorning.html` page if the time is before that

```
# Goodmorning before 6:00pm
RewriteCond %{TIME_HOUR} <18
RewriteRule ^time-test$ /goodmorning.html [R=302,L]
```

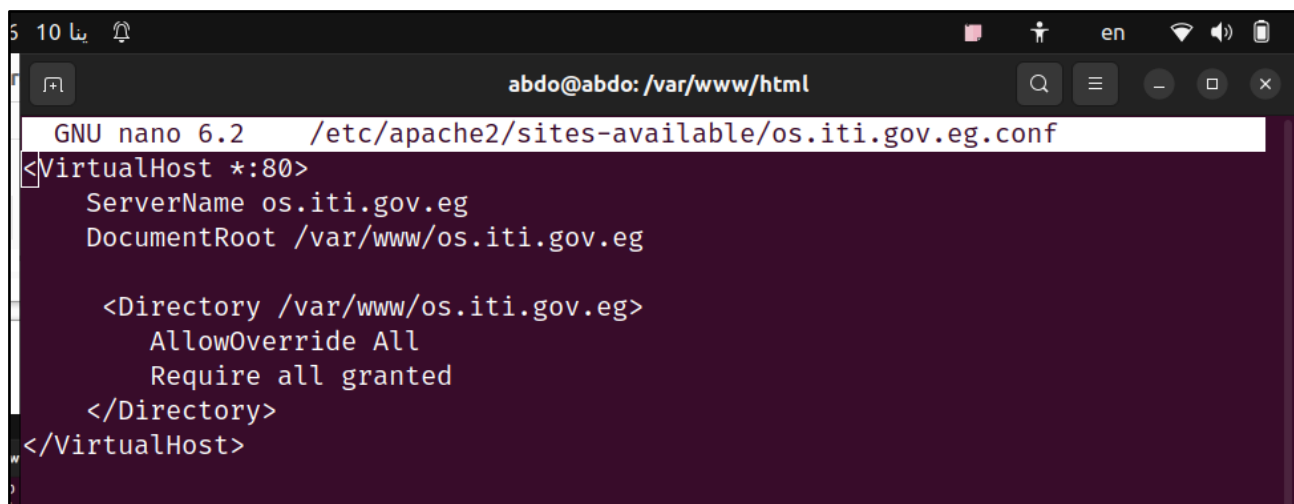
```
# Goodevening after 6:00pm
RewriteCond %{TIME_HOUR} >=18
RewriteRule ^time-test$ /goodevening.html [R=302,L]
```





4. Create a virtual host for os.iti.gov.eg to redirect to a local directory in your server

```
sudo mkdir -p /var/www/os.iti.gov.eg
sudo nano /var/www/os.iti.gov.eg/index.html
sudo nano /etc/apache2/sites-available/os.iti.gov.eg.conf
sudo a2ensite os.iti.gov.eg.conf
sudo systemctl reload apache2
```



```
abdo@abdo: /var/www/html
GNU nano 6.2 /etc/hosts *
#127.0.0.1 localhost
127.0.0.1 os.iti.gov.eg
127.0.1.1 abdo

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

